Percentile in Percentile: rating scholars against their career cohort by the impact of all their papers

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Abstract

Committees often compare scholars early in their careers, before citations have accumulated, yet citation counts and the h-index drift with field, calendar and seniority, and the h-index reads only one corner of a record. We propose PiP (percentile in percentile). Each paper is ranked against papers from the same year; for every quality threshold θ, the scholar's number of papers at or above θ is ranked against peers who started in the same year; PiP averages these standings over $\theta \geq \tau$. With $\tau = 0.5$, every paper in the top half of its year earns credit, a single top paper earns much, and weak papers earn nothing; τ is a dial between quantity and impact. We compute PiP for 11.1 million Semantic Scholar authors, using only information available at each evaluation year. Paper percentiles settle quickly (correlation 0.91 between ages 3 and 10), and PiP means the same in every cohort from 1960 to 2015, while the h-index at career age 10 rose by 40%. PiP predicts the next ten years of work better than the h-index in every setting and in each of 15 fields (at career age 10, Spearman 0.43 against 0.35 for the number of future top-10% papers). Indicators built from the same paper percentiles, such as top-10% counts or the sum or mean of percentiles, beat PiP on at most one of three outcomes. PiP also detects future high performers earlier, reaching the h-index's age-10 accuracy about two years sooner, and, measured at career age 5, it anticipates later academy memberships and fellowships better than the h-index (AUC 0.75 against 0.71). The results hold without self-citations and agree with Google Scholar profiles ($\rho = 0.78$).

Keywords: research evaluation; citation percentiles; h-index; career age; indicators; Semantic Scholar

1 Introduction

Hiring, promotion and award committees compare scholars with others at a similar stage of their careers. A committee deciding on tenure asks whether a candidate's record is strong for someone six years past the doctorate; a funding panel compares early-career applicants across departments. Bibliometric indicators enter these decisions alongside expert judgment, and the most common ones, citation counts and the h-index (Hirsch, 2005), are hard to read in this setting. Their values depend on the field, grow with seniority and drift with the calendar: the average h-index of a scholar ten years into their career is 40% higher for those who started in 2010 than for those who started in 1960 (Section 5.3). A raw value says little until it is compared with the values of comparable scholars.

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Many of these decisions are made early. Hiring and tenure committees, fellowship panels and early-career grants judge scholars three to eight years into their careers, when most of their papers are recent and few have had time to collect many citations. An indicator is most useful in this setting if it detects strong records early: if, measured at year five, it already separates the scholars who will go on to do important work from those who will not. Detecting performance early was one of the motivations for this work.

The h-index also reads only part of a record. It is the largest h such that h papers have at least h citations each: the corner where the sorted citation curve crosses the diagonal. Papers below the corner do not count, the citations of papers above it beyond h do not count, and a recent paper counts only once it has collected h citations, however well it compares with papers of its own age. A scholar with one landmark paper and a scholar with one average paper both have $h = 1$.

At the level of single papers, the field has settled on percentiles as the preferred normalization (Bornmann et al., 2013; Waltman, 2016; Hicks et al., 2015): a paper is compared with papers of the same publication year, and possibly the same field, by the share of them it out-cites. Percentiles do not depend on the skewed tail of the citation distribution, they need no assumption about how citations grow with age, and their meaning does not change with the calendar. Combining paper percentiles into an indicator for a scholar is less settled. Sums of percentiles, such as I3 (Leydesdorff and Bornmann, 2011), grow without bound with the number of papers, so many weak papers can outweigh one strong paper. Means ignore volume altogether. Counts of papers in the top 10% (Waltman et al., 2012; Bornmann and Marx, 2014) use one threshold and discard everything below it.

This paper proposes PiP, short for *percentile in percentile*, which places a scholar's papers in percentiles of their publication year and then places the scholar in percentiles of their career cohort. For each quality threshold θ, we count the scholar's papers at or above percentile θ and ask what share of peers, scholars who published their first paper in the same year, have fewer such papers. This standing, traced over all thresholds, forms a curve (Figure 1). PiP is the average height of the curve above a cutoff τ. Where the h-index reads one corner of a record, PiP measures the area under the whole curve. With $\tau = 0.5$, which we use throughout, every paper in the top half of its year earns credit, recent papers earn credit as soon as they stand out among papers of their own age, and a single paper in the top 1% earns credit at every threshold, including the highest ones, where most peers have nothing. Lowering τ moves weight towards the number of papers; raising it moves weight towards their impact.

We compute PiP from the Semantic Scholar Academic Graph (Kinney et al., 2023) for 11.1 million authors at every career age from 1 to 30, with peers and percentiles defined only from information available in the evaluation year. The paper makes four contributions:

1. **Definition and properties.** PiP is bounded, monotone in the addition or improvement of papers, gives nothing to papers below the cutoff, and factors into a productivity percentile times a quality multiplier (Section 4).

2. **Measurement over time.** Paper percentiles settle within three to five years, with field-specific windows (Section 5.1); the distribution of PiP is the same for every career cohort from 1960 to 2015 (Section 5.3); and a scholar's PiP is as stable over a career as their productivity and more stable than their citations or h-index (Section 5.4).

3. **Prediction.** PiP predicts a scholar's future top papers better than the h-index in every setting and field we test, whether the h-index is used raw or converted to a cohort percentile. Unlike counts of top papers or sums and means of paper percentiles, it predicts both how many top papers a scholar will publish and how good their papers will be (Section 5.5).

4. **Early detection.** Measured at the same career age, PiP identifies future high performers more accurately than the other indicators we test (the original PiP-AUC comes close), and it reaches the accuracy of the h-index at career age 10 about two years sooner (Section 5.6).

5. **Later recognition.** Measured early in a career, PiP anticipates later academy memberships, fellowships and prizes better than the h-index; for this purpose the impact end of the dial works best (Section 5.7).

6. **Robustness.** The results hold without self-citations; PiP rankings from Semantic Scholar agree with rankings from author-curated Google Scholar profiles; and a field-normalized version removes the advantage of high-citation fields without reordering scholars within a field (Sections 5.8–5.9).

An earlier version of this work (Tian and Ipeirotis, 2021) studied the predictability of paper and scholar percentiles on a smaller dataset of faculty at top US universities and rated scholars by the sum of their paper percentiles. Reviewers of that version objected to the sum, which rewards volume without bound, to the selected population, and to the lack of field-level results. This paper replaces the sum with PiP, moves to a population that includes everyone who ever published, and reports results by field. A first, asymmetric form of the score was described by Ipeirotis (2024) and is used by the website `https://www.pip-score.org`; Section 4 explains why we replace it.

2 Related work

The h-index and its variants. The h-index (Hirsch, 2005) combines productivity and impact in one number and has been studied, criticized and modified at length. The g-index (Egghe, 2006) gives credit to the citations above the corner; the m-quotient (Hirsch, 2005), the contemporary h-index (Sidiropoulos et al., 2007) and hI_a (Harzing et al., 2014) correct for career length by dividing by academic age or discounting old papers; Kaur et al. (2013) rescale the h-index by the field average. Most variants correlate highly with the original (Bornmann et al., 2011). They keep its basic construction, a count of papers above a citation threshold that depends on the scholar, and so keep its blindness to papers outside the core. The h-index is also inconsistent: adding the same paper to two scholars can reverse their order (Waltman and van Eck, 2012).

Percentiles. Percentile-based indicators compare each paper with a reference set of papers from the same year and field (Bornmann et al., 2013; Leydesdorff et al., 2011; Waltman and Schreiber, 2013; Bornmann and Williams, 2020). They are less sensitive to outliers than mean-based normalization (Schubert and Braun, 1986; Radicchi et al., 2008; Waltman and van Eck, 2019) and are recommended in the Leiden Manifesto (Hicks et al., 2015). Ties and the definition of the percentile need care; we use the share of the reference set with strictly fewer citations for scoring and Hazen's mid-rank (Hazen, 1914) for correlations. At the

level of a scholar, Bornmann and Marx (2014) propose counting papers in percentile classes and normalizing by academic age, and I3 (Leydesdorff and Bornmann, 2011; Rousseau, 2012) sums percentiles over a set of papers. Composite indicators such as the one of Ioannidis et al. (2019) combine several citation counts, and Abramo and D'Angelo (2014) weight field-normalized citations by inputs. PiP differs from these in normalizing twice: papers by their publication year, and the scholar by their career cohort, at every quality threshold.

Prediction of scholarly impact. Hirsch (2007) found that the h-index predicts a scholar's later h-index and citations better than counts of papers or citations do. Later work predicted future h-index values from features of a career (Acuna et al., 2012), questioned how much of that predictability is mechanical, since the future h-index contains the current one (Penner et al., 2013), and modeled impact as a stable individual parameter applied to papers of random quality (Sinatra et al., 2016); Clauset et al. (2017) review the area. We avoid the mechanical overlap by predicting outcomes built only from papers published after the evaluation year.

Citation windows and cumulative advantage. How long a paper needs before its citations reflect its impact depends on the field (Wang, 2013; Abramo et al., 2011). Citations also feed on themselves: well-known scholars receive more attention for the same work (Merton, 1968; Petersen et al., 2014), which affects any citation-based indicator and any test of its predictive validity. We return to this in Section 6.

3 Data

Papers and citations. We use the Semantic Scholar Academic Graph (Kinney et al., 2023), release of 27 October 2026, which is distributed under the ODC-BY license. It contains 226.0 million papers published from 1950 to 2025. For every paper we count citations by the year of the citing paper, which gives the cumulative citations at each age t (the number of years after publication), through calendar year 2025. These counts add up to 97.5% of the citation counts that Semantic Scholar reports for papers from 2000–2005, with a correlation of 0.999; the difference consists of citations from 2026 and from citing papers without a year.

Articles. Semantic Scholar contains many records that are not research articles. We keep the 71.5 million papers typed as journal articles, conference papers, reviews, studies, clinical trials, meta-analyses or case reports, and not as editorials, news or letters. Only these *articles* enter the reference sets, the paper counts and the h-index. In the 2000 cohort, 63% of all records are uncited at age 5, but only 30% of articles. Each article carries zero or more of Semantic Scholar's 23 fields of study.

Authors and careers. Semantic Scholar assigns author identifiers through automatic disambiguation. We link authors to their articles (277.8 million links) and define an author's *start year* s_a as the year of their first article, and their *career age* at evaluation year Y as $A = Y - s_a$. We follow every author from career age 1 to 30, or to 2025, whichever comes first. At each evaluation year an author is *active* if they have at least three articles and an h-index of at least three, both computed from citations received through Y. Because both

counts only grow, an author who becomes active remains active. In total, 11.1 million authors are active at some career age, giving 111.6 million author-age records. None of the quantities used at evaluation year Y depends on information after Y.

Google Scholar profiles. For validation we use 16,218 Google Scholar profiles collected for `https://www.pip-score.org`, mostly of faculty. Unlike Semantic Scholar's author records, these profiles are curated by the scholars themselves, so they are a check on automatic disambiguation (Section 5.9).

4 The PiP score

4.1 Paper percentiles

Fix an evaluation year Y. For an article j published in year $p_j \le Y$, let $c_j(Y)$ be the citations it has received through Y. Its *percentile* is the share of articles published in the same year with strictly fewer citations at Y:

$$y_j(Y) = \frac{\left|\{\, i : p_i = p_j,\ c_i(Y) < c_j(Y) \,\}\right|}{\left|\{\, i : p_i = p_j \,\}\right|} \in [0,1). \tag{1}$$

Uncited articles have $y_j = 0$. Every article is compared only with articles of the same publication year, evaluated at the same time, so y_j needs no correction for the age of the paper or for the growth of citations over the decades. A paper one year old competes only with other papers one year old. The main analysis pools all fields; Section 5.8 computes percentiles within fields.

4.2 Peers and standing curves

At evaluation year $Y = s_a + A$, the *peers* $\mathscr{P}$ of author a are the authors with the same start year who are active at Y. For a quality threshold $\theta \in [0,1]$, let

$$n_a(\theta) = \left|\{\, j \in a : p_j \le Y,\ y_j(Y) \ge \theta \,\}\right| \tag{2}$$

be the number of the author's articles at or above percentile θ; $n_a(0)$ is the number of articles. The author's *standing* at threshold θ is the share of peers with strictly fewer articles at or above θ:

$$g_a(\theta) = G_\theta(n_a(\theta)), \qquad G_\theta(m) = \frac{\left|\{\, b \in \mathscr{P} : n_b(\theta) < m \,\}\right|}{|\mathscr{P}|}. \tag{3}$$

The *standing curve* $\theta \mapsto g_a(\theta)$ starts at the author's standing on the number of articles and ends at their standing on the number of top articles. It need not be monotone: an author can rank higher among peers on the number of top-10% papers than on the number of papers (Figure 1).

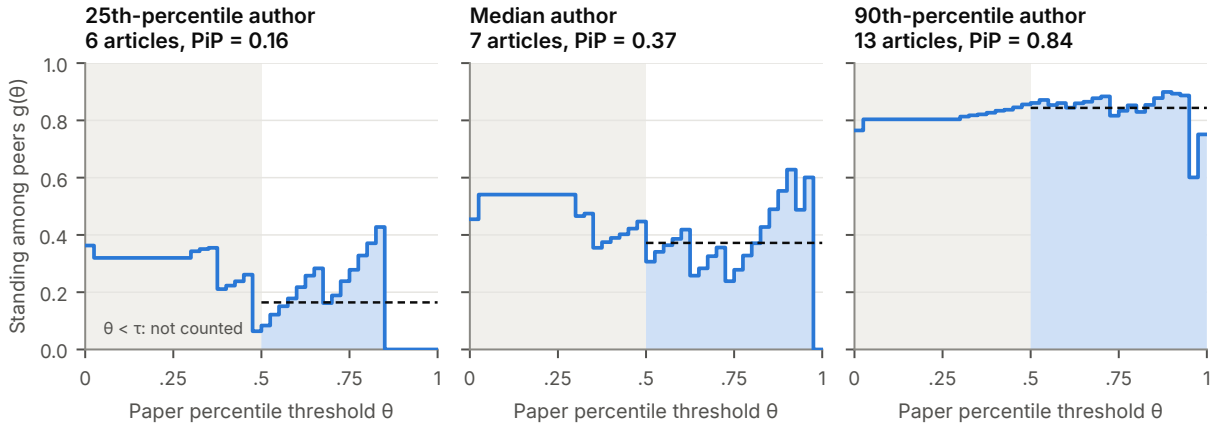


Figure 1: Standing curves of three authors who published their first article in 1995, evaluated in 2005 (career age 10). Each was chosen as the author whose PiP is closest to the 25th, 50th or 90th percentile of the cohort's scores. The curve gives, for each threshold θ, the share of the 1995 cohort's active authors who have fewer articles at or above percentile θ of their publication year. PiP is the shaded area to the right of $\tau = 0.5$, rescaled by $1/(1-\tau)$, or equivalently the average height of the curve there (dashed line). The median author has seven articles, two of them in the top 10% of their year and one in the top 5%. At thresholds from 0.85 to 0.95 this author stands above about half of the peers, who have fewer articles that highly cited.

4.3 Definition

The PiP score with cutoff $\tau \in [0,1)$ is the average standing over the thresholds at or above τ:

$$\mathrm{PiP}_\tau(a) = \frac{1}{1-\tau}\int_\tau^1 g_a(\theta)\,d\theta. \tag{4}$$

It is the area under the standing curve to the right of τ, rescaled to $[0,1]$. We use $\tau = 1/2$ throughout, so only articles in the top half of their publication year earn credit, and write PiP for $\mathrm{PiP}_{1/2}$. Section 5.2 shows how the cutoff moves the score between quantity and impact.

4.4 Properties

The following hold for a fixed peer group; we verified each numerically on simulated cohorts (the code is in the replication package).

Range. $0 \le \mathrm{PiP}_\tau(a) < 1$. The score is zero when, at almost every threshold above τ, no peer has fewer articles at or above that threshold than a, for example when a has no article above τ.

Monotonicity. Adding an article, or raising the percentile of an existing article, never lowers PiP_τ, because either change weakly increases $n_a(\theta)$ at every threshold and G_θ is non-decreasing. An article whose percentile stays below τ leaves the score unchanged.

No reward for volume alone. At $\tau = 1/2$, an author whose articles all fall below the median of their publication year scores zero, however many such articles they publish.

Credit for the best work. A single article in the top 1% raises $n_a(\theta)$ by one at every threshold, including the highest, where most peers have none. In a simulated cohort of 2,000 peers, a record of one article at the 99th percentile scores 0.45, while one article at the 50th percentile scores 0. The h-index cannot tell these two records apart (both have $h = 1$).

Factorization. Let $F(m) = |\{b \in \mathscr{P} : n_b(0) \le m\}|/|\mathscr{P}|$ be the productivity percentile, which is positive for every member of the peer group. Then

$$\mathrm{PiP}_\tau(a) = F(n_a(0)) \cdot Q_\tau(a), \qquad Q_\tau(a) = \frac{1}{1-\tau} \int_\tau^1 \frac{g_a(\theta)}{F(n_a(0))}\, d\theta. \tag{5}$$

The *quality multiplier* Q_τ compares the author's standing on counts of good articles with their standing on the number of articles. It exceeds one when the author ranks higher among peers on good articles than on volume. In our data this holds for 30–36% of active authors, depending on career age. Taking logarithms splits the variance of $\log \mathrm{PiP}_\tau$ across authors into a productivity part, a quality part and their covariance. For $\tau = 1/2$ and active authors who started in 1975–2005, the quality part accounts for 0.74–0.82 of the variance and the productivity part for about 0.30, with a small negative covariance (-0.04 to -0.11), at career ages 5, 10 and 20.

Consistency. PiP is not consistent in the sense of Waltman and van Eck (2012): adding the same article to two authors can reverse their order, because G_θ is not linear. In simulation, this happened in 0.7% of 3,000 random pairs. The h-index shares the property.

4.5 Relation to other indicators

Writing indicators in terms of the count curve $\theta \mapsto n_a(\theta)$ places PiP among existing ones. A top-x% count reads the curve at one threshold, $n_a(1-x)$, without comparing it with peers. I3 sums the percentiles, $\sum_j y_j = \int_0^1 n_a(\theta)\, d\theta$: the same integral without the peer comparison, so it grows without bound with the number of articles. The h-index reads one point of the analogous curve in citation space, where the number of articles with at least k citations equals k.

The first form of the score, which we call the *original PiP-AUC* (Ipeirotis, 2024), sorts an author's articles by percentile, $y_{(1)} \ge y_{(2)} \ge \cdots$, and computes the area under the step curve that plots $y_{(k)}$ against the productivity percentile $F(k)$:

$$\text{PiP-AUC}(a) = \sum_{k=1}^{n_a(0)} y_{(k)} \big[F(k) - F(k-1)\big] = \int_0^1 F(n_a(\theta))\, d\theta. \tag{6}$$

The second expression shows the difference from PiP: the original form compares the author's count at or above θ with peers' *total* counts rather than with their counts at or above θ. It is the product of $F(n_a(0))$ and

a weighted mean of the author's own percentiles, which keeps it bounded and monotone. But the weight of the k-th best article, $F(k) - F(k-1)$, is the share of peers with exactly k articles, which is zero for every k below the smallest article count among peers. Among the active authors of a cohort, nobody has fewer than three articles, so an author's best two articles receive no weight at all; among Google Scholar faculty profiles, the median author's best four articles receive none. The original form is therefore dominated by productivity: the productivity percentile accounts for 0.59–0.77 of the variance of its logarithm, and an author with one article at the 99th percentile scores zero, while an author with 40 weak articles scores 0.10 in the simulated cohort. A cutoff τ applied to the original form compares partial counts with growing total counts, and its median at career age 10 halves between the 1960 and 2010 cohorts. The symmetric comparison in PiP removes both problems.

4.6 Computation

We evaluate the integral on the grid $\theta_k = k/40$, $k = 0, \ldots, 39$, so that $\mathrm{PiP}_{1/2}(a) = \frac{1}{20} \sum_{k=20}^{39} g_a(\theta_k)$. For each start year and evaluation year, the distributions G_{θ_k} come from one pass over the peers' counts. The whole computation, 111.6 million author-age records, runs as a set of SQL queries in BigQuery.

5 Results

5.1 Paper percentiles settle within a few years

A scholar's PiP uses percentiles of recent papers, measured after only a few years of citations. We first check how well an early percentile anticipates the later one. For every article published from 1980 to 1995, we compute its percentile within its publication year at each age, using Hazen mid-ranks (Hazen, 1914), and correlate the percentiles at two ages within each publication year. Because the percentiles are ranks, this is the Spearman correlation of the citation counts. We report the average over publication years.

Percentiles settle quickly (Figure 2a, Table 1). The correlation with the percentile at age 10 is 0.72 after one year, 0.91 after three and 0.96 after five; with the percentile at age 30 it is 0.84 after three years and 0.89 after five. The ordinary least-squares slope of the later percentile on the earlier one is close to one (0.94 for ages 3 and 10), so early percentiles are not strongly biased towards the middle. Early percentiles have become more informative over time: the correlation between ages 1 and 10 was 0.72 for publication years before 1995 and 0.83 for 2015. For each cohort from 1960 to 1995, the correlation between ages 3 and 10 lies between 0.875 and 0.910.

The speed differs by field (Figure 2b), in line with Wang (2013) and Abramo et al. (2011). Three years give a correlation of about 0.9 with age 10 in biology (0.93), chemistry, physics, medicine and computer science (0.91), but 0.79 in history and 0.75 in art. The humanities and law need about five years to reach the agreement that the natural and life sciences reach in three. PiP uses percentiles at every age, including recent papers; Section 6 discusses what this means for scholars in slow fields.

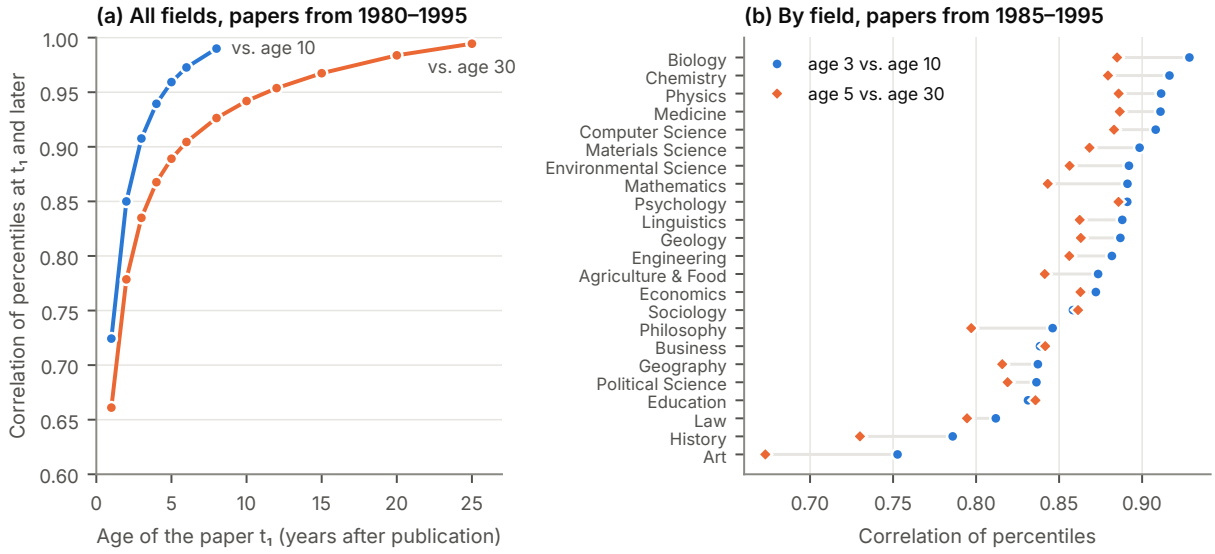


Figure 2: How well a paper's early citation percentile anticipates its later percentile. (a) Correlation between the percentile at age t_1 and at age 10 or 30, for 7.2 million articles published in 1980–1995, averaged over publication years. (b) The same correlation by field, for articles published in 1985–1995, with percentiles computed within the field and publication year; fields with more than 20,000 articles.

5.2 The cutoff τ is a dial between quantity and impact

The cutoff decides which papers earn credit. We computed PiP for $\tau \in \{0, 0.25, 0.5, 0.6, 0.75, 0.9\}$ for 3.43 million active authors at career age 10 (start years 1975–2010) and correlated each version, within cohort, with the author's number of articles, the mean percentile of their articles and their number of articles in the top 2.5% of their year (Figure 3a, Table 2).

As τ rises, the correlation with the number of articles falls from 0.86 to 0.37 and the correlation with the number of top-2.5% articles rises from 0.49 to 0.78. The correlation with mean quality peaks at $\tau = 0.75$. The original PiP-AUC sits at the quantity end of this range, closer to the article count (0.90) than even $\tau = 0$. In prediction, the same dial trades the *number* of future top-10% papers, best predicted at low τ, for their *quality*, best predicted at $\tau = 0.75$; the chance of a future top-1% paper is predicted about equally well from $\tau = 0.25$ to 0.6. Above $\tau = 0.6$ the score stops being comparable across cohorts: at $\tau = 0.9$ only four grid thresholds remain, 42% of recent authors score zero, and the median falls from 0.37 to 0.16 between the 1960 and 2010 cohorts.

We use $\tau = 0.5$ as the default. It has a one-line explanation (only papers in the top half of their year count), it balances the correlations with quantity (0.64) and with top papers (0.60), it is at the maximum for the top-1% outcome, and it is stable across cohorts. Users who care more about volume can lower it, and users who care more about impact can raise it up to about 0.6; the choice should be made before the evaluation and stated.

Holdout check. The default was chosen from results that pooled all cohorts, so we checked that it is not tuned to them. We split the same-age prediction settings by start year into a choice sample (1975–1990) and

Table 1: Agreement between a paper's citation percentile at an early and a later age (articles published 1980–1995, 7.2 million papers). Correlation of Hazen percentiles within publication year, the slope of the later on the earlier percentile, and the mean absolute change.

Ages	Correlation	Slope	Mean absolute change
1 vs. 10	0.724	0.846	0.161
2 vs. 10	0.850	0.911	0.115
3 vs. 10	0.908	0.944	0.086
5 vs. 10	0.959	0.975	0.051
3 vs. 30	0.835	0.882	0.125
5 vs. 30	0.889	0.918	0.100
10 vs. 30	0.942	0.956	0.068
20 vs. 30	0.984	0.989	0.030

Table 2: PiP at different cutoffs τ, career age 10. Columns 2–4: within-cohort Spearman correlation with characteristics of the record (3.43 million authors, start years 1975–2010). Columns 5–7: prediction of the next ten years (1.45 million authors, start years 1975–2000). Last column: median score of the 1960 and 2010 cohorts.

	Correlation with			Prediction			
τ	Articles	Mean pct.	Top 2.5%	Top-10% count	Top-1% AUC	Mean pct.	Median
0	0.856	0.214	0.491	**0.445**	0.766	0.223	0.44 → 0.40
0.25	0.776	0.329	0.537	0.443	0.770	0.265	0.43 → 0.39
0.5	0.637	0.466	0.602	0.427	**0.772**	0.321	0.41 → 0.38
0.6	0.573	0.506	0.633	0.415	0.771	0.340	0.41 → 0.37
0.75	0.473	**0.536**	0.688	0.388	0.766	**0.356**	0.38 → 0.33
0.9	0.370	0.486	**0.781**	0.347	0.750	0.338	0.37 → 0.16
Original PiP-AUC	0.897	0.131	0.421	0.434	0.756	0.190	0.29 → 0.26
h-index	–	–	–	0.352	0.719	0.230	–

a test sample (1991 and later). Applying the rule above to the choice sample alone (the highest AUC for a future top-1% paper among cutoffs whose distribution is stable across cohorts, $\tau \leq 0.6$) selects $\tau = 0.6$ at career age 10, with $\tau = 0.5$ within 0.001. In the test cohorts both cutoffs keep every comparison reported in Sections 5.5–5.6 at career ages 5, 10 and 15: they beat the h-index on all three outcomes, no other indicator beats either of them on more than one outcome, and both have the highest AUC for a future top-1% paper. Across τ from 0.25 to 0.6 that AUC varies by less than 0.01 in the test cohorts, so the exact choice in this range matters little for the outcome used to make it. Two differences between the samples are worth noting. The quantity–quality trade-off turns at a slightly higher cutoff in the older cohorts (between 0.6 and 0.75 rather than at 0.5), and in those cohorts at career age 5 the top-10% count ties $\tau = 0.5$ on the top-1% outcome and beats it on future quality, while $\tau = 0.6$ keeps its lead.

5.3 A PiP score means the same thing in every cohort

Committees compare people who started in different years, and indicators are often read against fixed rules of thumb. Figure 4 shows the distribution of PiP and of the raw h-index at career age 10 for every start year from 1960 to 2015. The PiP distribution hardly moves: the mean goes from 0.44 to 0.42, the lower quartile

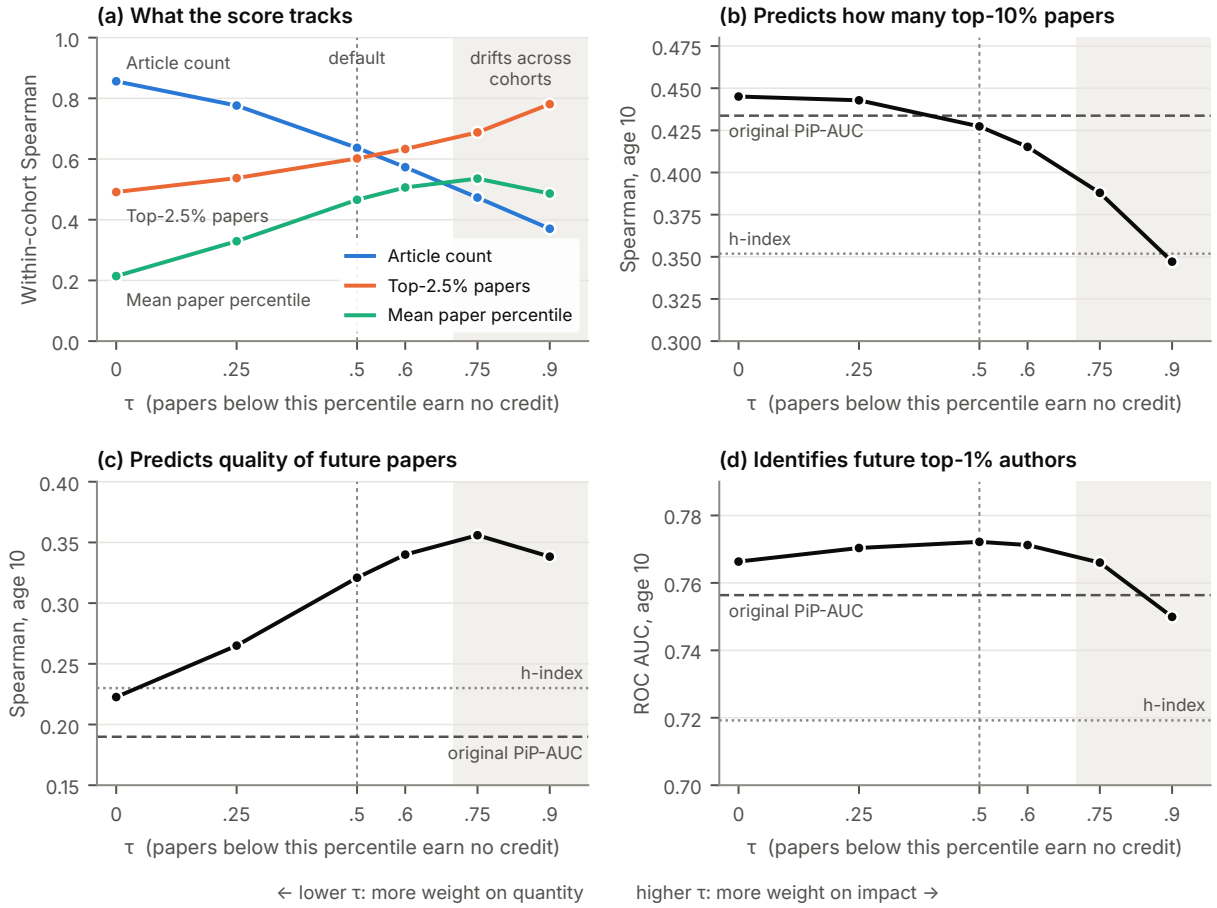


Figure 3: The cutoff τ moves PiP from quantity towards impact. Active authors at career age 10, start years 1975–2010. (a) Within-cohort Spearman correlation of PiP with the author's number of articles, number of articles in the top 2.5% of their year, and mean article percentile. (b–d) Prediction of the author's articles published in the next ten years (Section 5.5): (b) Spearman correlation with the number of future top-10% articles, (c) Spearman correlation with their mean percentile, (d) ROC AUC for publishing at least one top-1% article. Horizontal lines show the original PiP-AUC and the *h*-index. In the shaded range the score's distribution drifts across cohorts.

from 0.20 to 0.16, the upper quartile stays at 0.66–0.67 and the 90th percentile at 0.85. Over the same cohorts the mean *h*-index at age 10 rises from 4.4 to 6.2 and its 90th percentile from 7 to 11, following the growth of publication and citation rates. The cohorts themselves grow nearly a hundredfold, from 4,031 active authors in 1960 to 392,360 in 2015.

Stability of the distribution is not automatic. At high thresholds many peers tie at zero articles, and the share of ties changes with publication habits; the original form with a cutoff drifts (Section 4). In practice the stability means that a PiP of 0.6 at age 10 was as rare for the 1970 cohort as for the 2010 cohort, and that PiP values of people at the same career age can be compared directly, whatever their start year. Section 5.5 shows that this extends to comparisons across seniority.

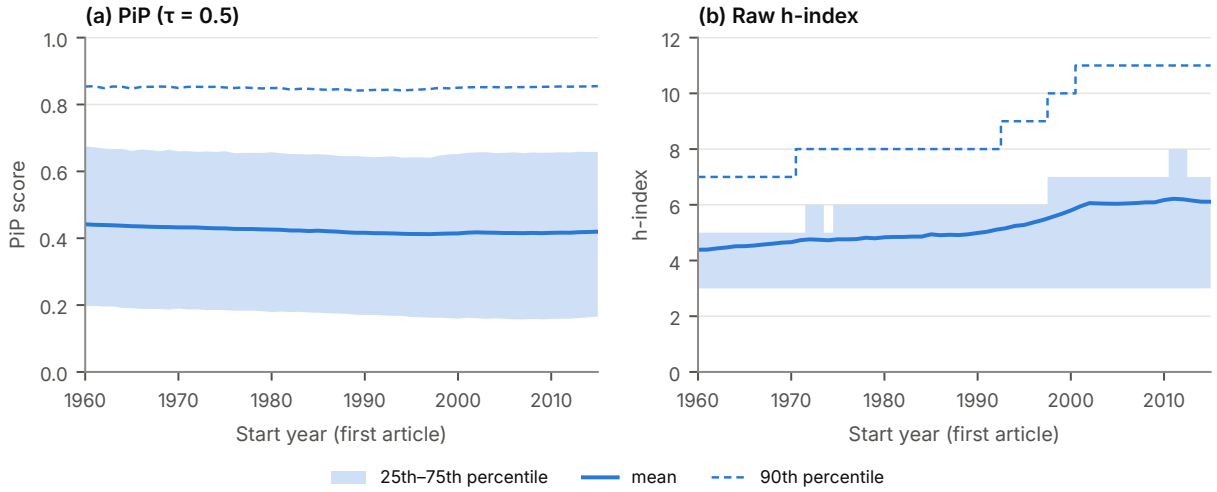


Figure 4: Distribution of (a) PiP ($\tau = 0.5$) and (b) the raw h-index of active authors at career age 10, by start year. The PiP distribution stays the same over 55 cohorts; the h-index grows by 40% on average.

Table 3: Stability of scholar indicators over a career. Correlation between an indicator's within-cohort percentile at two career ages (equal to the Spearman correlation of the raw values within cohort); active authors with start years 1975–1995. The 5-vs-30 comparison covers 434,000 authors.

Indicator	5 vs. 10	5 vs. 15	5 vs. 30	10 vs. 30	20 vs. 30
Articles	0.847	0.754	0.642	0.800	0.946
PiP ($\tau = 0.5$)	0.781	0.687	0.563	0.726	0.911
Original PiP-AUC	0.772	0.674	0.565	0.742	0.923
Citations	0.798	0.659	0.451	0.620	0.886
h-index	0.604	0.496	0.370	0.599	0.877

5.4 Stability over a career

An indicator that reshuffles scholars every few years is of little use for decisions with long consequences. For authors who started in 1975–1995, Table 3 correlates each indicator's within-cohort percentile at two career ages. PiP is about as stable as the original PiP-AUC and much more stable than the h-index: between ages 5 and 30 its correlation is 0.56, against 0.37 for the h-index and 0.45 for citations. Only the article count is more stable, as one would expect from a quantity that only accumulates. The h-index is particularly unstable early in a career, partly because many authors share the values 3 and 4.

5.5 Predicting future work

The most demanding test of an indicator meant for career decisions is whether it anticipates what the scholar does next. For each active author at evaluation year Y, we take the articles they publish in years $Y + 1$ to $Y + 10$ and score each at its own age 5, within its publication year. No article counts towards both the predictor and the outcome. We use three outcomes: the number of future articles in the top 10% (Spearman correlation), whether at least one future article is in the top 1% (ROC AUC), and the mean percentile of the future articles among authors with at least one (Spearman correlation).

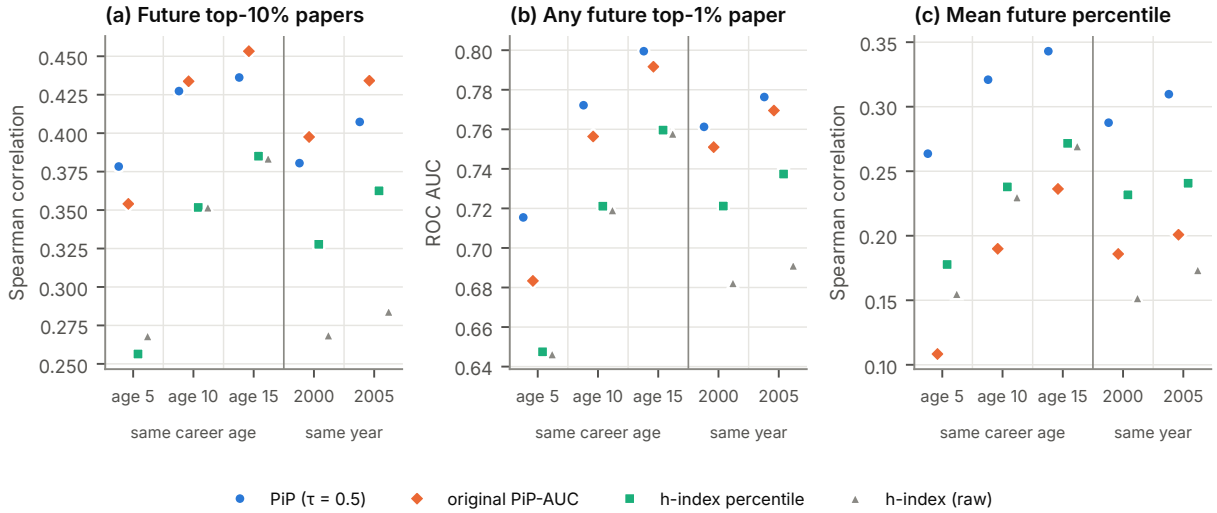


Figure 5: Prediction of the next ten years of a scholar's work from indicators at the evaluation year. Settings: active authors at career age 5, 10 or 15 (left of the vertical line), or all active authors in 2000 or 2005 at career ages 2–30 (right). Outcomes: (a) Spearman correlation with the number of future articles in the top 10% of their year; (b) ROC AUC for at least one future top-1% article; (c) Spearman correlation with the mean percentile of future articles. Each future article is scored at age 5.

Within one cohort, a raw indicator and its cohort percentile order people identically, so the comparison pools cohorts in two ways. *Same career age*: all active authors at career age 5, 10 or 15, from every start year since 1975 (the case of a tenure committee comparing candidates of similar seniority). *Same year*: all active authors in 2000 or in 2005, at career ages 2 to 30 (the case of a panel comparing people of different seniority). In the second setting we compare each indicator raw and as a percentile within its cohort.

PiP predicts better than the h-index on all three outcomes, in all five settings (Figure 5, Table 4). At career age 10, its correlation with the number of future top-10% papers is 0.43 against 0.35, its AUC for a future top-1% paper is 0.77 against 0.72, and its correlation with future quality is 0.32 against 0.23. The gap is larger when people of different seniority are compared: in 2005 the raw h-index reaches 0.28, 0.69 and 0.17. Converting the h-index into a cohort percentile helps (0.36, 0.74, 0.24) but does not close the gap. PiP gains much less from the conversion: in 2005 its cohort percentile reaches 0.42, 0.78 and 0.32, against 0.41, 0.78 and 0.31 for the raw score, because the raw score is already relative to the cohort and its distribution does not depend on the start year (Section 5.3). We report cohort percentiles for every indicator in the same-year settings.

Productivity and citations. The number of articles predicts the number of future top-10% papers as well as PiP or slightly better (0.44 against 0.43 at age 10), because productive authors keep producing; the original PiP-AUC, which is close to a productivity percentile, does the same. Neither predicts the quality of future work (correlations of 0.00 and 0.19). Total citations predict future quality about as well as PiP (0.30 against 0.32) but are weaker on the number of future top papers (0.29).

Table 4: Prediction of the next ten years of work. Career age 10 (1.45 million authors; raw values, which rank authors within one career age as their cohort percentiles do) and all active authors in 2005 (1.85 million authors, career ages 2–30; within-cohort percentiles, except where noted). Spearman correlation with the number of future top-10% articles, ROC AUC for at least one future top-1% article, and Spearman correlation with the mean percentile of future articles. The best value in each column is in bold.

	Career age 10			Year 2005, ages 2–30		
Indicator	Top-10%	Top-1%	Mean pct.	Top-10%	Top-1%	Mean pct.
PiP ($\tau = 0.5$)	0.427	**0.772**	0.321	0.416	**0.781**	0.318
Original PiP-AUC	0.434	0.756	0.190	0.433	0.770	0.201
h-index	0.352	0.719	0.230	0.363	0.737	0.241
raw, in 2005				0.284	0.691	0.174
Citations	0.290	0.714	0.298	0.312	0.733	0.303
Articles	0.437	0.740	0.000	**0.439**	0.757	0.019
Top-10% count	0.360	0.748	0.338	0.354	0.762	0.324
per year of career, raw				0.379	0.771	0.345
Top-1% count	0.250	0.641	0.228	0.262	0.698	0.225
I3 (sum of percentiles)	**0.438**	0.755	0.182	0.434	0.769	0.189
Mean percentile	−0.043	0.523	**0.394**	−0.042	0.526	**0.404**

Percentile-based indicators. The closest alternatives to PiP use the same paper percentiles but combine them differently: the number of papers in the top 10% (Waltman et al., 2012), also divided by career age (Bornmann and Marx, 2014); the number in the top 1%; I3, the sum of the percentiles (Leydesdorff and Bornmann, 2011); and their mean. Each is the best or nearly the best on one outcome and weak on another (Table 4). The mean percentile predicts the quality of future papers best (0.39 at age 10, against 0.32 for PiP), but says nothing about how many top papers will follow (−0.04) or whether one of them will be in the top 1% (AUC 0.52). I3 predicts the number of future top papers as well as the article count does (0.44) but not their quality (0.18). The top-10% count is the closest competitor: it predicts future quality slightly better than PiP (0.34 against 0.32) and the number of future top papers clearly worse (0.36 against 0.43). In all five settings, each of these indicators beats PiP on at most one of the three outcomes, and none beats it on the chance of a future top-1% paper.

Figure 6 places the indicators by how well they predict future quantity and future quality, with PiP at every cutoff. The cutoffs trace a frontier. With $\tau = 0$, PiP predicts at least as well as I3 and the original PiP-AUC on all three outcomes in every setting (with $\tau = 0.25$, in all but one); with $\tau = 0.75$ it does at least as well as the top-10% count in every setting; and from $\tau = 0.25$ to 0.75 it does better than the *h*-index on all three. Three indicators lie on or beyond the frontier. The mean percentile lies beyond it in the direction of pure quality, where it gives up any information about quantity. The article count lies just beyond it in two settings (at age 15 and in 2000), in the direction of pure quantity. The top-10% count per year of career, which differs from the plain count only when seniorities are mixed, lies on the curve next to $\tau = 0.75$. The cutoff therefore selects a point on a curve that matches or dominates the other ways of combining paper percentiles, and $\tau = 0.5$ sits near its knee: from $\tau = 0.25$ to 0.5 the curve gains 0.056 in future quality for 0.016 in quantity, and beyond 0.5 each gain in quality costs more in quantity.

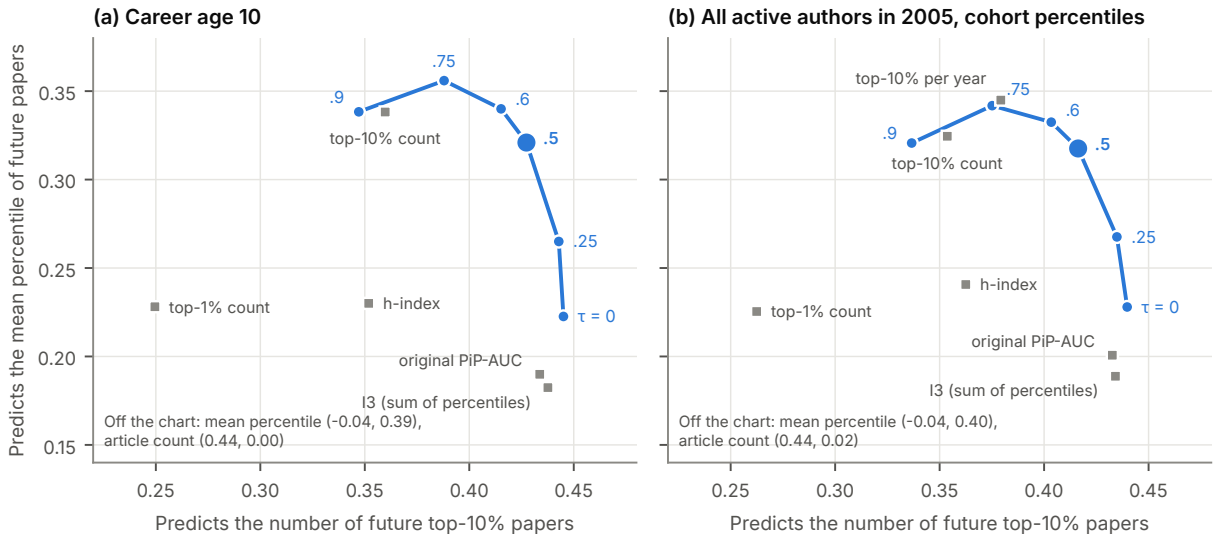


Figure 6: Indicators placed by how well they predict the number of a scholar's future top-10% papers (horizontal) and the mean percentile of their future papers (vertical), Spearman correlations. The blue curve is PiP at cutoffs from $\tau = 0$ to 0.9; the large dot is $\tau = 0.5$. Grey squares are indicators built from the same paper percentiles, the *h*-index and the original PiP-AUC. (a) Active authors at career age 10, raw values. (b) All active authors in 2005 at career ages 2–30, within-cohort percentiles (the top-10% count per year of career is raw). For the third outcome, a future top-1% paper, PiP at every cutoff from 0 to 0.75 has a higher AUC than every grey indicator in both settings.

Replication across cohorts. With more than a million authors per setting, sampling error is small: the standard error of a Spearman correlation of 0.4 is below 0.001. The more relevant question is whether the differences hold in different populations. We repeated the age-10 comparison separately for each start year from 1975 to 2000, 26 cohorts of 17,867 to 122,114 authors. PiP beats the *h*-index on all three outcomes in all 26 cohorts, by 0.06–0.10 on the number of future top papers. It beats the top-10% count on the number of future top papers (by 0.05–0.08) and on the top-1% outcome in all 26 cohorts, and falls behind it on future quality in 20 (by up to 0.07). Against I3, the article count and the original PiP-AUC, PiP wins on the top-1% outcome and on future quality in every cohort, and on the number of future top papers in 11 to 14 of the 26, with differences between -0.03 and $+0.04$. PiP is the best predictor of a future top-1% paper in every setting.

5.6 Early detection

The settings above compare indicators measured at one age. Early detection asks a different question: how soon does each indicator pick out the scholars who will do well later? To answer it we fix both the population and the outcome, and vary only the age at which the indicators are measured. The population is every author who is active at career age 3 (306,854 authors with start years 1975–2000); since activity persists, each of them has every indicator at every later age. The outcome is their work at career ages 11–20, scored as before (each article at its own age 5). We call an author a *future star* if they are in the top 10% of their start-year cohort, within the population, by the number of top-10% articles published at ages 11–20 (ties broken by the mean percentile of those articles). We then measure each indicator at career ages 3, 4, . . . , 10, and ask how

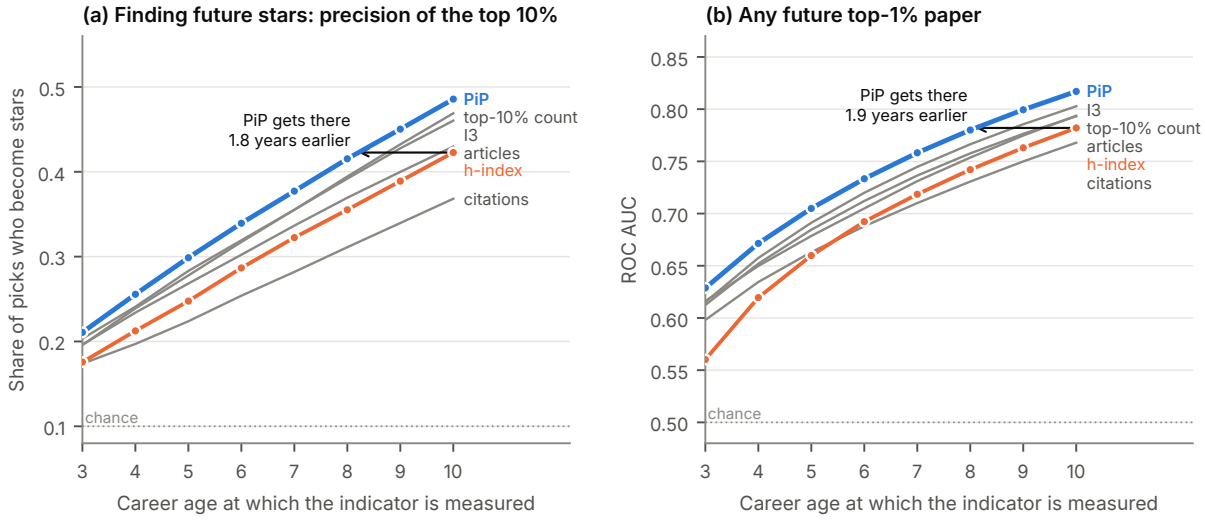


Figure 7: Early detection. Authors active at career age 3 (306,854; start years 1975–2000), with each indicator measured at career ages 3 to 10 and the outcome fixed: their articles published at career ages 11–20. (a) Of the authors in the top 10% of their cohort by the indicator, the share who become future stars (the top 10% of the cohort by the number of top-10% articles at ages 11–20). (b) ROC AUC for at least one top-1% article at ages 11–20. The arrow marks the career age at which PiP reaches the accuracy that the *h*-index has at age 10. The original PiP-AUC, not shown, lies just below PiP in both panels.

Table 5: How much earlier PiP reaches each indicator's accuracy, in years. For each target age T from 5 to 10, we find the (interpolated) age at which PiP first matches the indicator's accuracy at T; the table gives the range of T minus that age. Authors active at career age 3, outcome at career ages 11–20.

Compared with	Precision for future stars	AUC for future stars	AUC for a future top-1% paper
Citations	1.7–3.2	1.4–2.8	1.2–2.5
h-index	1.2–1.8	1.1–1.6	1.3–1.9
Articles	0.7–1.6	0.3–0.9	0.6–1.3
Top-10% count	0.5–0.6	1.0–1.3	0.8–1.3
I3	0.4–0.7	0.2–0.4	0.4–0.8
Original PiP-AUC	0.0–0.2	0.0–0.2	0.3–0.4

well it finds the future stars: the precision of picking the top 10% of each cohort by the indicator (chance is 0.10), and the AUC for the same outcome. We also report the AUC for publishing at least one top-1% article at ages 11–20. Each indicator enters as its percentile within the start-year cohort.

PiP is the most accurate indicator at every measurement age on both AUC measures, and on precision at every age but 9 and 10, where the original PiP-AUC matches it to within 0.001 (Figure 7). Measured at career age 3, the top 10% of a cohort by PiP contains 21% future stars, twice the chance rate, against 18% for the *h*-index; at age 5 the shares are 30% and 25%, and at age 10, 49% and 42%. The AUC for a future top-1% paper is 0.63 against 0.56 at age 3 and 0.71 against 0.66 at age 5. Because every indicator improves with age, the gaps translate into time (Table 5): PiP reaches the accuracy that the *h*-index has at age 10 about two years earlier, at age 8, and the accuracy of the citation count up to three years earlier. Its lead over the article count, the top-10% count and I3 is smaller, at most 1.6 years, and the original PiP-AUC is close behind (within 0.4

Table 6: ROC AUC for holding a senior honor. Columns 2–5: indicators measured at career age 5 or 10 in Semantic Scholar, honors received later; all active authors (the comparison uses every honoree and a 10% random sample of the others), only people with a Wikidata item, and only our Google Scholar profiles matched to Semantic Scholar (mostly faculty). Columns 6–7: current Google Scholar scores of 15,890 profiled scholars against holding an honor and having an English Wikipedia article (agreement, not prediction). PiP's bootstrap intervals are about ± 0.015 for all authors and ± 0.02 within the smaller groups; differences between indicators are estimated more precisely because they are paired.

	Predicting later honors (Semantic Scholar)				Google Scholar, now	
Future honorees / positives	All, age 5 1,064	All, age 10 1,555	Wikidata, 10 751	GS faculty, 10 890	Honors 1,497	Wikipedia 2,160
PiP ($\tau = 0.5$)	0.749	0.816	0.628	0.631	0.712	0.716
PiP ($\tau = 0.75$)	**0.760**	**0.826**	**0.646**	**0.638**	0.711	**0.721**
Original PiP-AUC	0.719	0.791	0.603	0.622	0.696	0.692
h-index	0.710	0.782	0.610	0.623	0.708	0.704
Citations	0.701	0.789	0.629	0.627	**0.714**	**0.721**
Articles	0.693	0.772	0.569	0.611	0.677	0.675
Top-10% count	0.747	0.816	0.644	0.634	0.709	0.720
I3	0.721	0.788	0.595	0.622	0.696	0.692
Mean percentile	0.527	0.568	0.570	0.535	0.611	0.611

years). The results are the same for the larger population of authors active at age 5 (711,986 authors): there PiP leads the *h*-index by 1.2–1.8 years and the citation count by 1.6–2.9 years.

On the other two outcomes of Section 5.5 the ordering at each age matches the one found there: the article count predicts the number of future top-10% papers slightly better than PiP at the earliest ages, and the mean percentile predicts their average quality best. PiP's advantage in early detection is in finding the scholars at the top, who combine many papers with highly cited ones, and in anticipating exceptional papers.

5.7 Later recognition

The tests so far judge citation indicators by later citations. As a check against an outcome that is not a citation count, we ask whether the indicators anticipate later scientific honors. We take honors from Wikidata: 65 senior honors given for research, namely membership of national academies (such as the US National Academy of Sciences, the Royal Society and the Leopoldina), fellowships of learned societies (AAAS, ACM, IEEE, APS and others) and major prizes and fellowships (such as the Guggenheim, MacArthur and Gairdner awards). We leave out plain society memberships, state orders, student scholarships and the citation-based Clarivate Citation Laureates. Honors reach Semantic Scholar authors in two ways: through Wikidata items that carry a Semantic Scholar author ID (41,272 items), and through items that carry a Google Scholar ID (12,434 honorees), which we link to Semantic Scholar with the title matching of Section 5.9. We measure each indicator at career age 5 or 10, as a percentile within the start-year cohort (start years 1960–2000), and drop people whose first dated honor came at or before that age; most honors come decades later. This leaves 1,064 future honorees at age 5 and 1,555 at age 10.

Among all active authors, PiP measured at career age 5 picks out the future honorees with an AUC of 0.75, against 0.71 for the *h*-index and 0.70 for citations; at age 10, 0.82 against 0.78 and 0.79 (Table 6). The paired bootstrap intervals for PiP's lead over the *h*-index exclude zero at both ages ($+0.039$ [0.029, 0.049]

and +0.035 [0.030, 0.040]). PiP ties the top-10% count (+0.000 [−0.005, 0.005] at age 10), and the impact end of the dial does better than the default: at $\tau = 0.75$ the AUC is 0.76 and 0.83. Recognition rewards highly cited work more than volume: the article count, I3 and the original PiP-AUC, which lean towards volume, do worst apart from the mean percentile.

Within groups that are already notable, every indicator separates future honorees only weakly (AUC 0.59–0.65). There, PiP is ahead of the *h*-index at career age 10 (by 0.018 [0.010, 0.025] among people with a Wikidata item and 0.008 [0.003, 0.014] among the Google Scholar faculty) and level with it at age 5, while the top-10% count and PiP at $\tau = 0.75$ or higher do best. Restricting to honors with a recorded date gives the same pattern. On the Google Scholar profiles, where scores and honors are measured at the same time, the indicators are close: PiP is slightly ahead of the *h*-index for holding an honor (+0.004 [0.001, 0.008]) and for having a Wikipedia article (+0.012 [0.009, 0.015]), and total citations and the top-10% count do as well or better.

The recognition test therefore supports PiP over the *h*-index with an outcome outside citation counts, most clearly for detecting future honorees among all scholars and early in their careers. It also shows that the default cutoff is not the best choice for every purpose: for anticipating recognition, an evaluator should move the dial towards impact. Two caveats apply. Wikidata covers honors unevenly across countries and fields, and honors depend on nominations and networks as well as on the work, so these AUCs understate how well any indicator tracks merit.

5.8 Fields

We assign each author the most frequent field among their articles by career age 10 (whole counting) and repeat the analyses by field. Fifteen fields have at least 3,000 active authors with start years 1975–2000. In every one of them, PiP predicts better than the *h*-index on all three outcomes; it predicts future top-1% papers and future quality better than the original PiP-AUC in every field, while the original form is better on the number of future top-10% papers in about half of them, as it is overall. For example, in economics PiP reaches 0.39 / 0.74 / 0.27 on the three outcomes against 0.31 / 0.67 / 0.17 for the *h*-index, and in mathematics 0.38 / 0.75 / 0.28 against 0.29 / 0.68 / 0.19.

Pooling fields in the paper percentiles favors fields with high citation rates. A field-neutral score would put 10% of each field's authors in the top 10% of their cohort; with pooled percentiles the share ranges from 1.4% in education and 2.1% in economics to 14.7% in physics (Figure 8). We therefore also compute a field-normalized PiP, in which a paper's percentile is computed within its publication year and the author's main field (when the paper carries that field; otherwise the pooled percentile is kept), and peers are restricted to authors with the same main field and start year. For 1.45 million authors at career age 10, this brings every field's share of the cohort top 10% to between 6.9% and 11.6%. Within a field the ranking barely changes: the correlation between the pooled and the field-normalized score is 0.91–0.99 per field. The remaining spread comes from papers outside the author's main field, which keep their pooled percentile.

The two versions answer different questions. With outcomes measured on pooled percentiles, the pooled score predicts best (0.427 / 0.772 / 0.321 against 0.414 / 0.770 / 0.298); with field-normalized outcomes, the field-normalized score predicts best (0.407 / 0.772 / 0.316 against 0.405 / 0.769 / 0.300). Both remain far

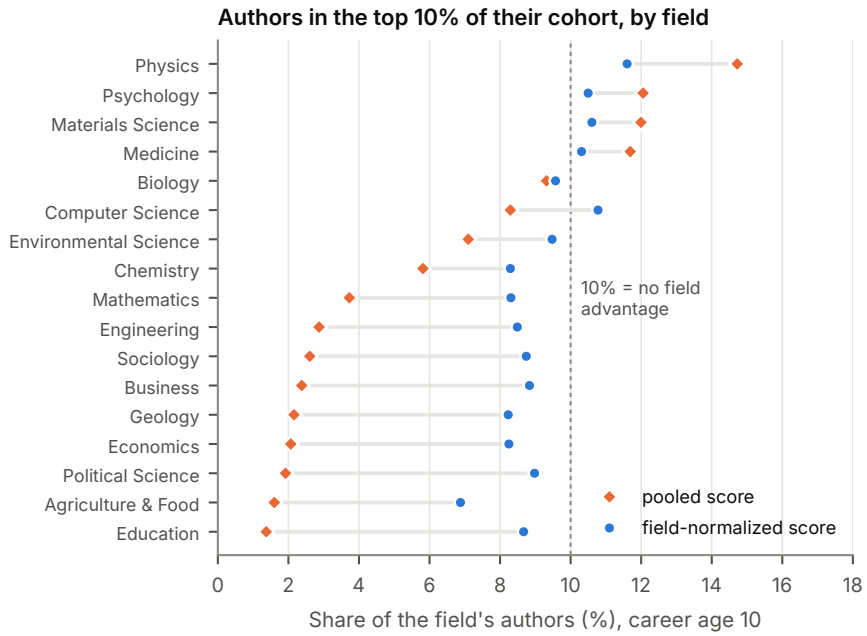


Figure 8: Share of each field's authors who are in the top 10% of their start-year cohort at career age 10, with PiP computed from pooled percentiles and with the field-normalized version. Fields with at least 5,000 authors.

ahead of the *h*-index (0.333 / 0.717 / 0.209 on field-normalized outcomes). For comparisons across fields we recommend the field-normalized version; within a field the two are nearly interchangeable.

5.9 Robustness

Self-citations. A citation counts as a self-citation when the citing and the cited paper share at least one author. These are 8.1% of all citations; such rates depend strongly on the definition and the citation window (Aksnes, 2003). We removed them and reran the whole pipeline: percentiles, peer groups (the *h*-index threshold is then applied without self-citations), scores and outcomes (Table 7 in the appendix). The predictions of the number of future top-10% papers and of a future top-1% paper move by at most 0.003 for PiP. The *h*-index loses slightly more (0.352 to 0.340 at age 10). The prediction of future quality drops by about 0.02 for every indicator, which shows that part of the persistence of quality runs through authors citing their own work. The ordering of the indicators does not change.

Author disambiguation. Semantic Scholar's author records come from automatic disambiguation, which can split one person across several records or merge different people. We matched the 16,218 Google Scholar profiles to Semantic Scholar through their publications: a profile's work matches a Semantic Scholar paper on its normalized title (at least 25 characters) and a publication year within one year, and the profile's counterpart is the Semantic Scholar author with the same surname who holds the most matched works. We find a confident counterpart (at least five matched works and at least 30% of the profile's titles) for 14,511 profiles (89%). For the median person, 76% of their Google Scholar works are found in Semantic Scholar, and 90%

of those found sit under the main author record. Splitting is common, though: 64% of people have at least two of their works under a second record with the same surname and first initial, and 42% under a third. The main record is smaller than the profile (a median 67% of the profile's cited works, and 78% of its *h*-index), and its start year is later than the profile's for 47% of people, because early work is missing or sits under another record.

Despite these differences, the two sources rank people similarly. For the 9,796 people who have scores in both sources (Semantic Scholar start year 1995 or later, evaluated in 2025), the Spearman correlation between sources is 0.82 for the *h*-index, 0.78 for the original PiP-AUC and 0.78 for PiP; each source computes percentiles against its own population (all active Semantic Scholar authors, and Google Scholar profiles). The agreement is lower for people split across three or more records (0.73) than for people with one record (0.76), but not by much. We read this as evidence that the results above are not artifacts of one disambiguation algorithm, and as a warning that individual Semantic Scholar scores can be off when a person's work is split.

Survivorship. The population includes everyone who ever published, not only those who stayed in research. Whether an author is still publishing at career ages 11–15 is predicted by their age-5 indicators with an AUC of 0.68 for the article count, 0.65 for the original PiP-AUC, 0.62 for PiP, 0.58 for the *h*-index and 0.54 for citations. Continuing to publish is a matter of volume, so the indicators that weigh volume most predict it best. The prediction results of Section 5.5 count authors who stop publishing as having no future top papers, which favors indicators that predict survival; PiP leads on the top-1% and quality outcomes despite this.

6 Discussion

Why PiP improves on the *h*-index. The *h*-index summarizes a record by one point; PiP uses every paper above the cutoff, through a comparison with peers at every quality level. Three properties explain most of the gain in prediction. First, PiP gives credit to the best papers in proportion to how rare they are among peers: a single top paper raises the standing at the highest thresholds, where most peers have none, and the *h*-index ignores everything above the corner. Second, it gives credit to recent papers as soon as they stand out among papers of the same age, while the *h*-index waits until they have accumulated enough raw citations; this matters most early in a career, when most papers are recent, and is the main source of PiP's lead in early detection (Section 5.6). Third, its scale does not depend on the cohort, so scholars of different seniority can be compared, while the raw *h*-index favors those who started later, when citation rates were higher, and those who have been active longer.

Prediction and cumulative advantage. Citations reward visibility as well as quality: well-known scholars attract more citations for comparable work (Merton, 1968; Petersen et al., 2014). Part of the predictive power of any citation-based indicator, PiP and the *h*-index alike, may therefore come from reputation feeding itself rather than from the persistence of research quality. Removing self-citations, the most mechanical channel, leaves the ranking of indicators unchanged and lowers the prediction of future quality by only about 0.02; it does not rule out the broader effect of reputation, which would need an external source of variation to

separate. Since the effect applies to every indicator we compare, it does not explain the differences between them. The test against later honors (Section 5.7) uses an outcome that is not a citation count and gives the same ordering of PiP and the h-index, although honors are themselves shaped by visibility.

Fields and windows. The pooled score favors fields with high citation rates, and the field-normalized version removes most of that advantage. Field normalization brings its own choices: the field scheme, the assignment of multi-field papers and the assignment of an author to a main field (Zitt et al., 2005; Waltman and van Eck, 2015, 2019). Percentiles of recent papers are also noisier in fields where citations accumulate slowly (Section 5.1). In those fields, an evaluation might count only papers older than two or three years, at the cost of ignoring the most recent work.

Limitations.

- *Disambiguation.* Split author records shorten careers and move start years, which places the affected person in a later cohort with a partial record. For evaluating individuals, a curated record such as a Google Scholar or ORCID profile should be used.
- *Whole counting.* Every co-author receives full credit for every paper, regardless of the number of authors or their roles. A fractional version would weight each paper's contribution to $n_a(\theta)$ by the author's share (Waltman and van Eck, 2015); we have not tested it.
- *The peer group.* Peers are defined by the year of the first article. Authors who start late, interrupt their careers or move between research and other work are compared with a cohort whose careers ran differently. The activity filter (three articles and an h-index of three) also excludes the earliest stages of a career, where PiP is not defined.
- *Reference set.* The article reference set depends on Semantic Scholar's publication types, which are incomplete; untyped records are excluded from both the reference sets and the counts.
- *Consistency.* PiP is not consistent in the sense of Waltman and van Eck (2012), although reversals are rare in simulation.

Responsible use. PiP is meant to inform comparisons among scholars at a similar career stage, alongside reading the work itself and expert judgment, as the Leiden Manifesto (Hicks et al., 2015) and DORA (San Francisco Declaration on Research Assessment, 2012) recommend. Its cutoff should be chosen before an evaluation, not tuned to favor a candidate, and the field version should be used when people from different fields are compared. A predictive correlation of 0.4 leaves most of a scholar's future unexplained.

7 Conclusion

PiP rates a scholar by the citation percentiles of all their papers above a cutoff, compared with peers who started in the same year at every quality level. On 11.1 million Semantic Scholar authors it means the

same thing for every cohort since 1960, is as stable over a career as productivity, and predicts future top papers and future quality better than the *h*-index in every setting and field we tested. Among indicators built from the same paper percentiles, it is the one that predicts both how many top papers a scholar will publish and how good their papers will be, and it detects future high performers about two years earlier than the *h*-index. A single parameter moves it between quantity and impact, along a frontier that matches or dominates counting, summing or averaging paper percentiles, except for the mean percentile at the pure-quality end. The construction needs only publication years and citation counts by year, and the code computes it at full scale in a database; a public implementation on Google Scholar profiles is available at `https://www.pip-score.org`.

Data and code availability

The SQL that builds every table in this paper, the numerical checks of the properties in Section 4, the data behind every figure and the plotting scripts are in the `paper/` directory of `https://github.com/ipeirotis-org/scholar_v2`. Semantic Scholar data are available under the ODC-BY license from the Semantic Scholar Datasets API.

Competing interests

Panos Ipeirotis developed and maintains `https://www.pip-score.org`, a free, non-commercial website that computes the original PiP-AUC from Google Scholar profiles. The authors have no other competing interests.

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References

Abramo, G., Cicero, T., and D'Angelo, C. A. (2011). Assessing the varying level of impact measurement accuracy as a function of the citation window length. *Journal of Informetrics*, 5(4):659–667.

Abramo, G. and D'Angelo, C. A. (2014). How do you define and measure research productivity? *Scientometrics*, 101(2):1129–1144.

Acuna, D. E., Allesina, S., and Kording, K. P. (2012). Future impact: Predicting scientific success. *Nature*, 489(7415):201.

Aksnes, D. W. (2003). A macro study of self-citation. *Scientometrics*, 56(2):235–246.

Bornmann, L., Leydesdorff, L., and Mutz, R. (2013). The use of percentiles and percentile rank classes in the analysis of bibliometric data: Opportunities and limits. *Journal of Informetrics*, 7(1):158–165.

Bornmann, L. and Marx, W. (2014). How to evaluate individual researchers working in the natural and life sciences meaningfully? A proposal of methods based on percentiles of citations. *Scientometrics*, 98(1):487–509.

Bornmann, L., Mutz, R., Hug, S. E., and Daniel, H.-D. (2011). A multilevel meta-analysis of studies reporting correlations between the h index and 37 different h index variants. *Journal of Informetrics*, 5(3):346–359.

Bornmann, L. and Williams, R. (2020). An evaluation of percentile measures of citation impact, and a proposal for making them better. *Scientometrics*, 124:1457–1478.

Clauset, A., Larremore, D. B., and Sinatra, R. (2017). Data-driven predictions in the science of science. *Science*, 355(6324):477–480.

Egghe, L. (2006). Theory and practise of the g-index. *Scientometrics*, 69(1):131–152.

Harzing, A.-W., Alakangas, S., and Adams, D. (2014). hIa: An individual annual h-index to accommodate disciplinary and career length differences. *Scientometrics*, 99(3):811–821.

Hazen, A. (1914). Storage to be provided in impounding municipal water supply. *Transactions of the American Society of Civil Engineers*, 77(1):1539–1640.

Hicks, D., Wouters, P., Waltman, L., De Rijcke, S., and Rafols, I. (2015). Bibliometrics: the leiden manifesto for research metrics. *Nature*, 520(7548):429–431.

Hirsch, J. E. (2005). An index to quantify an individual's scientific research output. *Proceedings of the National Academy of Sciences*, 102(46):16569–16572.

Hirsch, J. E. (2007). Does the h index have predictive power? *Proceedings of the National Academy of Sciences*, 104(49):19193–19198.

Ioannidis, J. P. A., Baas, J., Klavans, R., and Boyack, K. W. (2019). A standardized citation metrics author database annotated for scientific field. *PLOS Biology*, 17(8):e3000384.

Ipeirotis, P. (2024). The PiP-AUC score for research productivity. Blog post, `https://www.behind-the-enemy-lines.com/2024/01/the-pip-auc-score-for-research.html`.

Kaur, J., Radicchi, F., and Menczer, F. (2013). Universality of scholarly impact metrics. *Journal of Informetrics*, 7(4):924–932.

Kinney, R., Anastasiades, C., Authur, R., Beltagy, I., Bragg, J., et al. (2023). The Semantic Scholar Open Data Platform. arXiv:2301.10140.

Leydesdorff, L. and Bornmann, L. (2011). Integrated impact indicators compared with impact factors: An alternative research design with policy implications. *Journal of the American Society for Information Science and Technology*, 62(11):2133–2146.

Leydesdorff, L., Bornmann, L., Mutz, R., and Opthof, T. (2011). Turning the tables on citation analysis one more time: Principles for comparing sets of documents. *Journal of the American Society for Information Science and Technology*, 62(7):1370–1381.

Merton, R. K. (1968). The Matthew effect in science. *Science*, 159(3810):56–63.

Penner, O., Pan, R. K., Petersen, A. M., Kaski, K., and Fortunato, S. (2013). On the predictability of future impact in science. *Scientific Reports*, 3:3052.

Petersen, A. M., Fortunato, S., Pan, R. K., Kaski, K., Penner, O., Rungi, A., Riccaboni, M., Stanley, H. E., and Pammolli, F. (2014). Reputation and impact in academic careers. *Proceedings of the National Academy of Sciences*, 111(43):15316–15321.

Radicchi, F., Fortunato, S., and Castellano, C. (2008). Universality of citation distributions: Toward an objective measure of scientific impact. *Proceedings of the National Academy of Sciences*, 105(45):17268–17272.

Rousseau, R. (2012). Basic properties of both percentile rank scores and the I3 indicator. *Journal of the American Society for Information Science and Technology*, 63(2):416–420.

San Francisco Declaration on Research Assessment (2012). San Francisco Declaration on Research Assessment. `https://sfdora.org/read/`.

Schubert, A. and Braun, T. (1986). Relative indicators and relational charts for comparative assessment of publication output and citation impact. *Scientometrics*, 9(5-6):281–291.

Sidiropoulos, A., Katsaros, D., and Manolopoulos, Y. (2007). Generalized Hirsch h-index for disclosing latent facts in citation networks. *Scientometrics*, 72(2):253–280.

Sinatra, R., Wang, D., Deville, P., Song, C., and Barabási, A.-L. (2016). Quantifying the evolution of individual scientific impact. *Science*, 354(6312):aaf5239.

Tian, S. and Ipeirotis, P. (2021). On the predictability of utilizing rank percentile to evaluate scientific impact. arXiv:2103.16025.

Waltman, L. (2016). A review of the literature on citation impact indicators. *Journal of Informetrics*, 10(2):365–391.

Waltman, L., Calero-Medina, C., Kosten, J., Noyons, E. C. M., Tijssen, R. J. W., van Eck, N. J., van Leeuwen, T. N., van Raan, A. F. J., Visser, M. S., and Wouters, P. (2012). The Leiden Ranking 2011/2012: Data collection, indicators, and interpretation. *Journal of the American Society for Information Science and Technology*, 63(12):2419–2432.

Waltman, L. and Schreiber, M. (2013). On the calculation of percentile-based bibliometric indicators. *Journal of the American Society for Information Science and Technology*, 64(2):372–379.

Waltman, L. and van Eck, N. J. (2012). The inconsistency of the h-index. *Journal of the American Society for Information Science and Technology*, 63(2):406–415.

Waltman, L. and van Eck, N. J. (2015). Field-normalized citation impact indicators and the choice of an appropriate counting method. *Journal of Informetrics*, 9(4):872–894.

Waltman, L. and van Eck, N. J. (2019). Field normalization of scientometric indicators. In *Springer handbook of science and technology indicators*, pages 281–300. Springer.

Wang, J. (2013). Citation time window choice for research impact evaluation. *Scientometrics*, 94(3):851–872.

Zitt, M., Ramanana-Rahary, S., and Bassecoulard, E. (2005). Relativity of citation performance and excellence measures: From cross-field to cross-scale effects of field-normalisation. *Scientometrics*, 63(2):373–401.

A Self-citations

Table 7: Prediction results with and without self-citations (with / without). Self-citations are removed throughout: in the paper percentiles, the peer groups, the scores and the outcomes.

Setting	Indicator	Top-10% count	Top-1% AUC	Mean pct.
Age 5	PiP	.378 / .378	.715 / .712	.264 / .234
	Original PiP-AUC	.354 / .355	.683 / .681	.108 / .093
	h-index	.268 / .257	.646 / .640	.155 / .148
Age 10	PiP	.427 / .427	.772 / .771	.321 / .298
	Original PiP-AUC	.434 / .435	.756 / .755	.190 / .171
	h-index	.352 / .340	.719 / .714	.230 / .221
Age 15	PiP	.436 / .437	.799 / .799	.343 / .325
	Original PiP-AUC	.453 / .454	.792 / .790	.236 / .219
	h-index	.384 / .374	.758 / .754	.270 / .261
Year 2005	PiP	.407 / .407	.776 / .777	.310 / .291
	Original PiP-AUC	.434 / .437	.770 / .771	.201 / .188
	h-index	.284 / .277	.691 / .689	.174 / .169

B Prediction results in all settings

Table 8: Prediction of the next ten years of work in all five settings: Spearman correlation with the number of future top-10% articles / ROC AUC for at least one future top-1% article / Spearman correlation with the mean percentile of future articles. Same-year settings include career ages 2–30.

Setting	PiP ($\tau = 0.5$)	Original PiP-AUC	h-index (raw / cohort pct.)	
Age 5	.378 / .715 / .264	.354 / .683 / .108	.268 / .646 / .155	.256 / .647 / .178
Age 10	.427 / .772 / .321	.434 / .756 / .190	.352 / .719 / .230	.352 / .721 / .238
Age 15	.436 / .799 / .343	.453 / .792 / .236	.384 / .758 / .270	.385 / .760 / .272
Year 2000	.381 / .761 / .288	.398 / .751 / .186	.269 / .682 / .152	.328 / .721 / .232
Year 2005	.407 / .776 / .310	.434 / .770 / .201	.284 / .691 / .174	.363 / .737 / .241